

Portugal Administrative GIS & Freguesia Population Dataset

Project Overview

The project visualizes official geographic boundaries in Portugal both The Nomenclature of Territorial Units for Statistics (NUTS) and Official Administrative Charter of Portugal (Municipality and Freguesia; CAOP) and combined it with population data at the Freguesia level. The final outputs are prepared as GIS-ready datasets and professional cartographic products.

The project integrates:

- NUTS 1, NUTS 2 and NUTS 3 statistical regions
- Local administrative level boundaries
- Freguesia-level population data

Project Objectives

1. Building a clear administrative and statistical geographic hierarchy for Portugal.
2. Preparing NUTS 1, NUTS 2 and NUTS 3 datasets.
3. Processing local administrative CAOP boundaries.
4. Integrating Freguesia population data and boundaries.
5. Performing spatial and attribute-level QA/QC.
6. Producing clean and deliverable GIS-ready Shapefile.
7. Generating professional maps for visualization and documentation.

Data Processing Workflow

1. Downloaded NUTS and CAOP datasets from Eurostat GISCO and Dgterritorio
2. Filtered the NUTS and CAOP datasets for Portugal
3. Separated NUTS 1, nuts 2 AND nuts 3 Layers
4. Merged CAOP layers of different region in Portugal
5. Downloaded the INE Population dataset.
6. Inspected geographic identifiers and attribute fields.
7. Matched population records to Freguesia geographic identifiers
8. Calculated and verified geographic attributes
9. Aggregated the population data in Freguseia level
10. Join population data to the Freguesia boundary
11. Created population-based symbology.
12. Performed QA/QC validation.
13. Exported final deliverable GIS datasets
14. Produced cartographic layouts and documentation

Tools

QGIS: Select features, Merge vector layers, Aggregate, Join, Symbology, Attribute table, Vector geometry, Check validity, QA/QC, Layout

Microsoft Word: Documentation

Microsoft Excel: Data Dictionary

Data Sources

Eurostat regional level NUTS datasets were obtained from official European geographic data sources GISCO 2024. On the other hand, local administrative datasets were obtained from CAOP 2025. However, Population data with reference year 2021 were integrated at the freguesia level boundaries. Therefore, 2021 population represented using 2025 Freguesia administrative boundaries

The datasets with geographic reference are listed below:

Datasets	Geographic level	Usage
Eurostat GISCO NUTS 2024 ¹	NUTS 1–3	Regional hierarchy
CAOP 2025 ²	Municipality and Freguesia	Administrative boundaries
Population dataset - INE Census 2021 ³	Freguesia	Population attributes

NUTS: Year 2024, 1:1 Million, EPSG:3035 — ETRS89 / LAEA Europe
Map Projection Grid: EPSG:4326 (WGS 84)

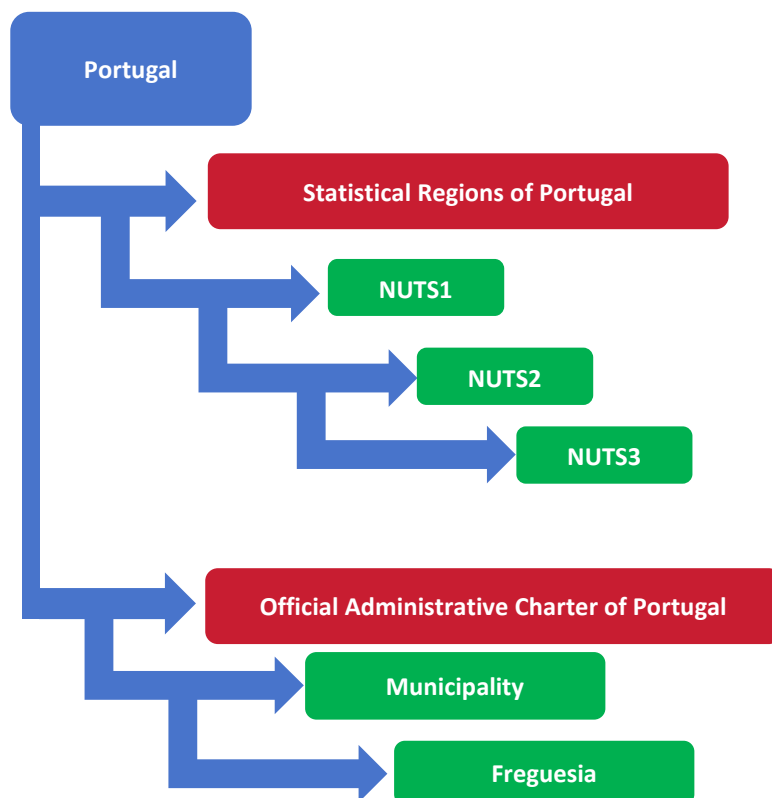
¹ <https://ec.europa.eu/eurostat/web/gisco/geodata/statistical-units/territorial-units-statistics>

² <https://www.dgterritorio.gov.pt/atividades/cartografia/cartografia-tematica/caop>

³ <https://mapas.ine.pt/download/index2021LugaresFregs.phtml>

Geographic Hierarchy

Portugal's geographic structure includes both territorial and local administrative units . The NUTS classification is used for statistical and regional level, while municipalities and Freguesias represent local administrative level geography.



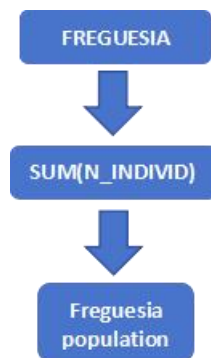
Geographic levels used in this project are listed below:

Level	Geography	Role
Country	Portugal	National level
NUTS 1	Territory of the Mainland and Autonomous Regions	Statistical classification
NUTS 2	Administrative Regions	Statistical and planning
NUTS 3	Inter-municipal Communities and Metropolitan Areas	Detailed regional statistics
Municipality	Município	Local administrative level
Freguesia	Freguesia	Local administrative subdivision
NUTS 1, 2 and 3 are statistical subdivisions by Eurostat Municipality and Freguesia are local administrative levels.		

Final Freguesia Population Dataset

Population data were joined to the freguesia administrative boundaries using geographic identifiers. Population data contained multiple records associated with the same freguesia were aggregated with sum function. Then, the relevant population values were being joined to the freguesia boundary layer. The final population field represents the 2021 population associated with the corresponding freguesia.

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The identifiers of attribute table are listed below:

Field	Description
GEO_ID	Freguesia geographic identifier
FREGUESIA	Freguesia name
MUNICIPIO	Municipality
DISTRICT	District
NUTS1	NUTS 1 region
NUTS2	NUTS 2 region
NUTS3	NUTS 3 region
AREA_HA	Area in hectares
POPULATION	Total number of inhabitants
COUNTRY	Country
SOURCE	Boundary/source information

QA/QC Results

After performing Quality control the following result has been found:

Check	Result
Total Freguesia features	3,259
Missing GEO_ID	0
Duplicate GEO_ID	0
Invalid geometries	0
Geometry errors	0
Population NULL	302

Population coverage is incomplete because the population data source does not use exactly the same Freguesia geographic configuration as the administrative boundaries. As a result, 302 current Freguesia records could not be directly matched to a population value. These records were retained with NULL population values rather than assigning potentially incorrect values.

Limitations

- Administrative boundaries and population data have different reference years.
- Some freguesias do not have corresponding population records in the population datasets.